

MUHAMMAD SUFYAN

✉ mohammadsufyan577@gmail.com

🔗 SufyanRx

🌐 sufyann

EDUCATION

Joint European Masters in Nuclear Physics

Thesis Defence Sep'26

Universidad de Sevilla | Université de Caen Normandie | Università degli Studi di Catania | Universidad Complutense de Madrid

14.3/20 (Mention Bien)

Department of Nuclear Physics and Applications

Master's thesis in progress: "Towards Automation of Quality Assurance in Radiotherapy Clinical Trials in Head and Neck Cancer Patients | Analytical Approaches in Dosimetry | Basics of Radiotherapy | Physics of Medical Devices | Machine Learning and Data Analysis | Computational Numerical Methods | Physics For Therapy | Medical Physics | Accelerator Physics & Application | Experiments in Nuclear Physics

Bachelors of Science in Physics

February 2024

COMSATS University Islamabad

3.54/4.00 (Excellent)

Physics Department, CIIT

Computational Methods in Physics | Quantum Mechanics | Mathematical Methods in Physics | Statistical Mechanics | Atomic & Molecular Physics | Computational Material Sciences | Nuclear Physics | High Energy Physics

RESEARCH EXPERIENCE

Automating QA in Radiotherapy Clinical Trials in Head and Neck Cancer Patients

01 Feb'26 - Continued

Prof. Dr. Juliette Thariat | Laboratoire de Physique Corpusculaire de Caen, (LPC, Caen)

Skills Gained: Python, PyTorch, Pydicom, Multimodal Data Handling

- Investigating Tumor Topology Complexity (TTC) to address a key limitation in radiotherapy QA, where treatment plans are labeled as pass/fail without distinguishing whether failure arises from suboptimal planning or intrinsic complex anatomy. This work aims to quantify anatomical and dosimetric complexity using multi-institutional DICOM data (CT, RTSTRUCT, RTDOSE) and integrate it into multimodal learning frameworks for more interpretable and fair QA evaluation.
- Conducted a 5-day research visit to MIT Marburg Ion-Beam Therapy Center under the [HITRIplus \(European Heavy Ion Therapy Research Integration Plus\)](#) program, with Prof. Juliette Thariat (Centre François Baclesse), gaining exposure to proton and carbon ion therapy workflows and treatment planning systems.

Toxicity Prediction in Head and Neck Cancer Patients using Machine Learning

15th Sep'25 - 15th Jan'26

Prof. Dr. Juliette Thariat | Laboratoire de Physique Corpusculaire de Caen, (LPC, Caen)

Skills Gained: Python, LightGBM, Scikit-learn

- Developed and validated a LightGBM pipeline using clinical tabular data to predict xerostomia, mucositis, and dysphagia in 684 head & neck patients, evaluated with an 80/20 train-test split and stratified 3-fold cross-validation for robust, reproducible performance assessment. Check at : [Internship Report](#)

Digital Twins in Radiotherapy: Beam Comparison and Plan Optimization with MatRad

15th July'25 - 14th Sep'25

Prof. Dr. Juliette Thariat | Centre François Baclesse, Caen

Skills Gained: MatRad, DICOM (CT, RTSTRUCT, RTDOSE), DVH analysis

- We developed digital twin simulations from anonymized patient DICOM datasets to evaluate photon, proton, helium, and carbon ion treatments. We optimized radiotherapy plans in MatRad to improve conformity and homogeneity indices, and analyzed DVHs for organs at risk and tumor coverage.

Plasma Diagnostics Internship at SMART Tokamak

27th Jan'25 - 28th Feb'25

Prof. Dr. Eleonora Viezzer | SMART Tokamak, Seville

Skills Gained: Plasma Experimentation, Impedans Measurements, Langmuir Probe

- I worked as an intern with the SMART Tokamak team, focusing on measuring plasma parameters such as potentials, ion and electron currents, electron temperature, and densities using a Langmuir probe. Our team installed, tested, and analyzed the probe to ensure accurate plasma diagnostics.

Final Year Project on "Tau Decays as a Probe of New Physics"

Feb'23 – Feb'24

Prof. Dr. Farida Tahir | COMSATS University, Islamabad

Skills Gained: FormCalc, FeynArts, Analytical Calculations, Particle Physics Modeling

- We explored Tau's pure and semi-leptonic decays in Standard Model as well as in R-Parity Violating Supersymmetry under the supervision of Prof. Dr. Farida Tahir. Analyzed parameters like branching ratios, CKM matrix entries, lepton universality, and investigated New Physics limits in the R-parity violating supersymmetric Model.

LABORATORY EXPERIENCE

- **Deep learning for particle physics classification:** Implemented multi-layer neural network in PyTorch for binary classification of high-energy physics events (7,600-sample dataset), including custom Dataset/DataLoader implementation, comprehensive model evaluation (accuracy, precision, recall, AUC-ROC), Data Augmentation and Optuna Optimization.
- **Radiation dose simulation and shielding analysis with RayXpert:** Modeled a contaminated effluent tank facility with complex piping geometry, assigned radioactive sources (volume and surface distributions), performed Monte Carlo dose calculations, and generated spatial dose mapping for radiation protection assessment and shielding optimization.
- **Beta spectroscopy using Geiger-Müller counters and SiPMs:** efficiency and attenuation analysis.
- **Gamma spectroscopy with scintillation detectors (NaI, CsI, LYSO, BGO):** energy calibration and comparison.

SKILLS AND LANGUAGES

Medical Physics Tools: MatRad, Treatment Planning & DVH Analysis, RT-DICOM Handling, Radiotherapy QA

Programming and ML: Python, PyTorch, Pydicom, LightGBM, Scikit-learn, SHAP Analysis, Linux, $\LaTeX$

Data Analysis & Experimental: Spectroscopy, Pulse Processing, RayXpert

Organizational: Event Management, Team Management, Public Speaking

Languages: English[C1], French[A1], Urdu[NS], Punjabi[NS]

HONORS AND AWARDS

Normandie Region Scholarship

Sep'24 - Sep'25

Regional Government of Normandie & the University of Caen,

- Awarded a scholarship to pursue my Joint European Masters in Nuclear Physics.

STFS - Science Talent Forming Scheme Scholar

Sep'18 - Feb'24

Pakistan Science Foundation

- Recognized as one of the potential young scientists in the country. Funded a project titled, "Removal of Pb^{+2} from water using **Citrus Jhimbiri** as bio-adsorbent". Also, awarded a fully funded scholarship for intermediate and Bachelor's education, including a final year research project.

Most Dedicated Volunteer

15th July - 25th July'23

Pakistan Young Innovative Minds' Society

- For Managing transport for the 36th IYPT at Murree and guiding Team Brazil among delegates from 17 countries.

Silver Honorer

5th July'23

International Astronomy and Astrophysics competition

- Participated in this international competition and was awarded this honor for my performance.

SCHOOLS & TOURNAMENTS ATTENDED

CaeSAR: Caen Strategy for Accelerating Research, Pont-l'Évêque, France: Participant	3 - 5 Sep '25
LHC School: 12th School on LHC Physics, Islamabad: Participant	22nd Aug – 01st Sep'23
IYPT: International Young Physicists' Tournament, Murree: Local Organizing Committee	19th - 24th July '23
ISPAD: International School of Physics and Allied Sciences, Islamabad: Participant	13 - 17 March '23

SYNERGISTIC AND LEADERSHIP ACTIVITIES

Director of Registrations: Pakistan Young Physicists' Tournament, Islamabad	19th Oct'23 - Present
Office Assistant: Pakistan Young Innovative Minds' Society, Islamabad	01 Oct'22 - 01 Oct'23
Co-Founder: E-Physics Academy, Pakistan	13th Feb'22 - Feb'23
Juror: Pakistan Young Physicists' Tournament, Pakistan	Nov'21 - Sep'24
Community Education Awareness Lead: United Nations Girls' Education Initiative, Pakistan	Nov'13 - Dec'14

REFERENCES

Prof. Dr. Juliette Thariat

Professor of Radiation Oncology

Centre François Baclesse; Université de Caen Normandie; LPC Caen (CNRS/IN2P3)

Caen, France

Email: j.thariat@baclesse.unicancer.fr

Relationship: Current Thesis Supervisor; Previous Internship Supervisor; Course Instructor (Basics of Radiotherapy)

Dr. Aurélien Corroyer-Dulmont

Head of Artificial Intelligence Department

Centre François Baclesse, AImaging group, ISTCT Unit 6030 CNRS

Caen, France

Email: a.corroyer-dulmont@baclesse.unicancer.fr

Relationship: Future PhD Co-Supervisor